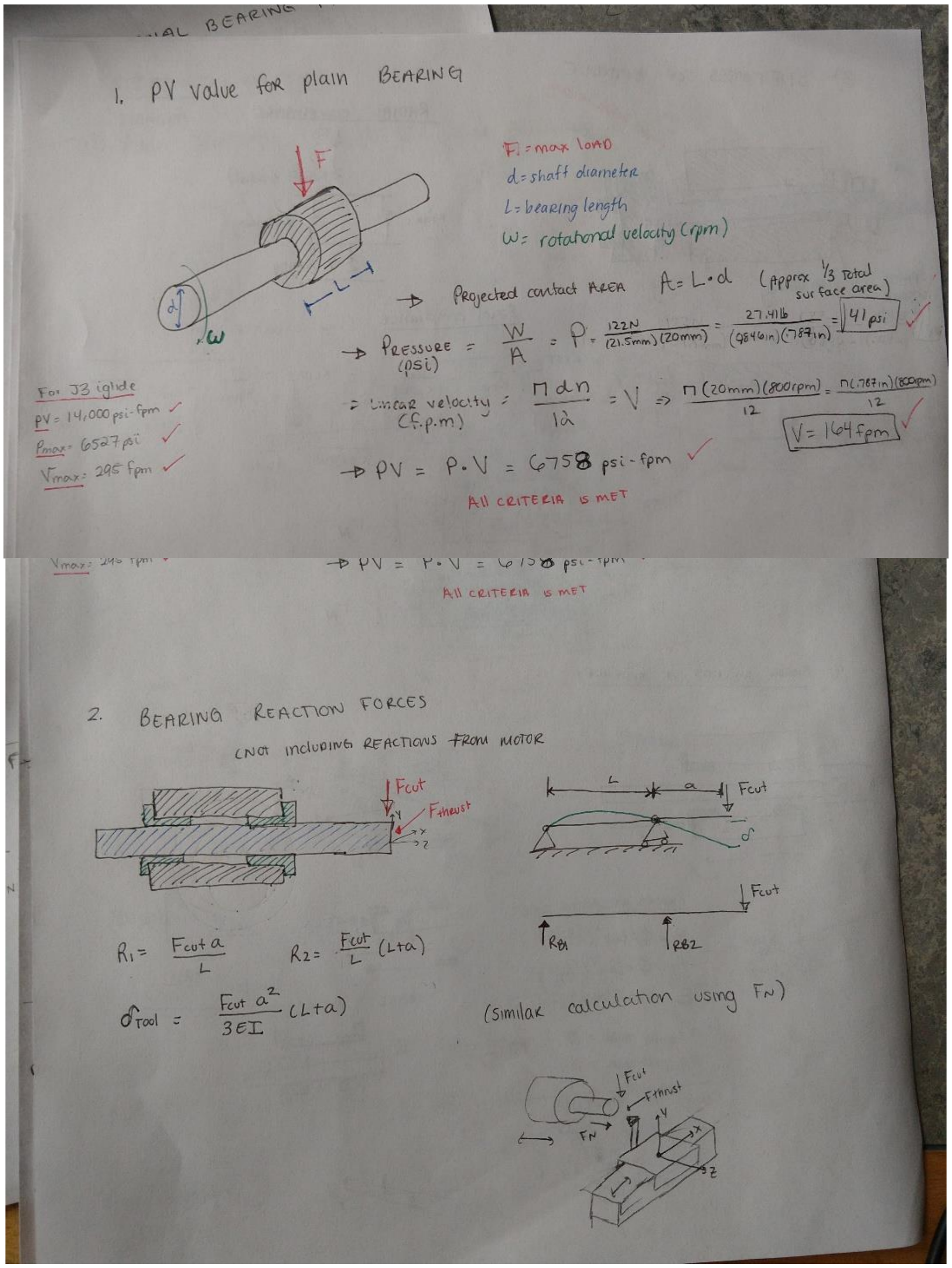
Rachel Hoffman
Week 10 Write-Up
4/14/2018

7. Appendix: Calculations



1. PV value for plain BEARING

F = max load
d = shaft diameter
L = bearing length
ω = rotational velocity (rpm)

→ Projected contact AREA $A = L \cdot d$ (Approx 1/3 total surface area)

→ Pressure (psi) $= \frac{W}{A} = P = \frac{122N}{(21.5mm)(20mm)} = \frac{27.41lb}{(.846in)(.787in)} = \boxed{41 psi}$

→ Linear velocity (f.p.m) $= \frac{\pi d n}{12} = V \Rightarrow \frac{\pi (20mm)(800rpm)}{12} = \frac{\pi (.787in)(800rpm)}{12}$

$\boxed{V = 164 fpm}$

→ $PV = P \cdot V = 6758$ psi-fpm

All CRITERIA IS MET

For J3 iglide
PV = 14,000 psi-fpm
P_{max} = 6527 psi
V_{max} = 295 fpm

2. BEARING REACTION FORCES
(NOT including REACTIONS FROM MOTOR)

$$R_1 = \frac{F_{cut}\, a}{L} \qquad R_2 = \frac{F_{cut}}{L}(L+a)$$

$$\delta_{Tool} = \frac{F_{cut}\, a^2}{3EI}(L+a)$$

(similar calculation using F_N)

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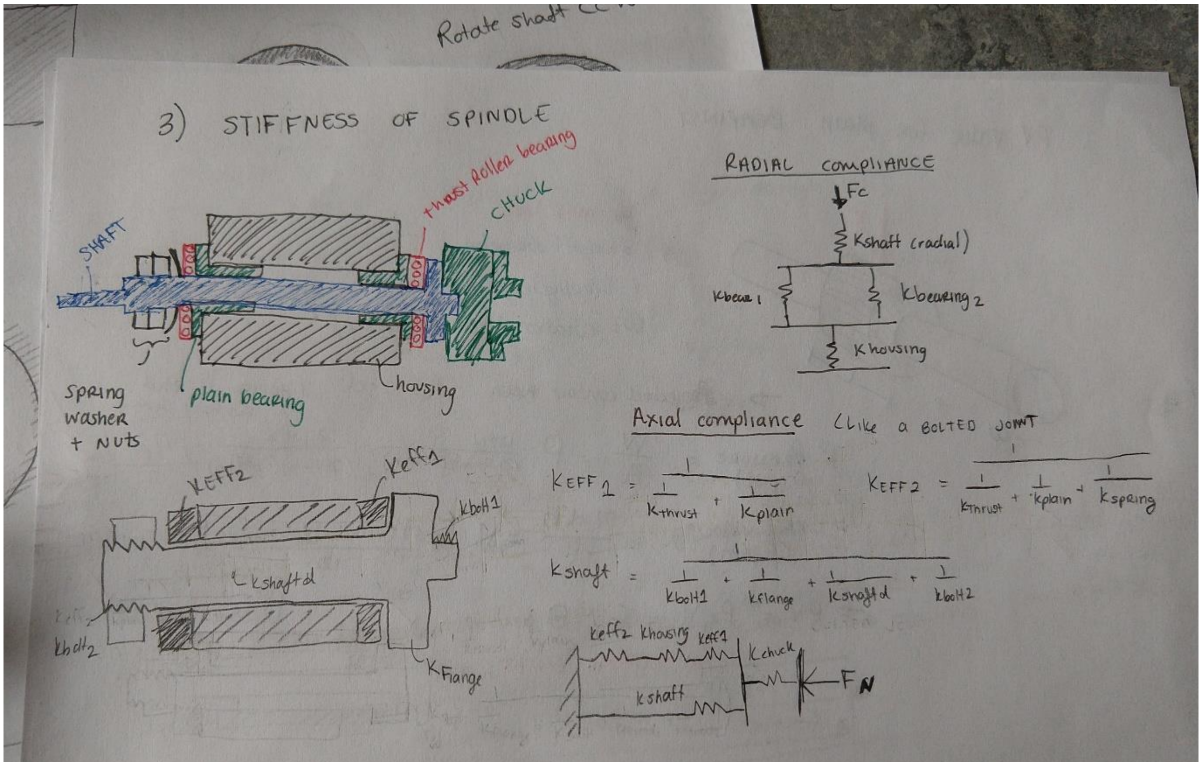
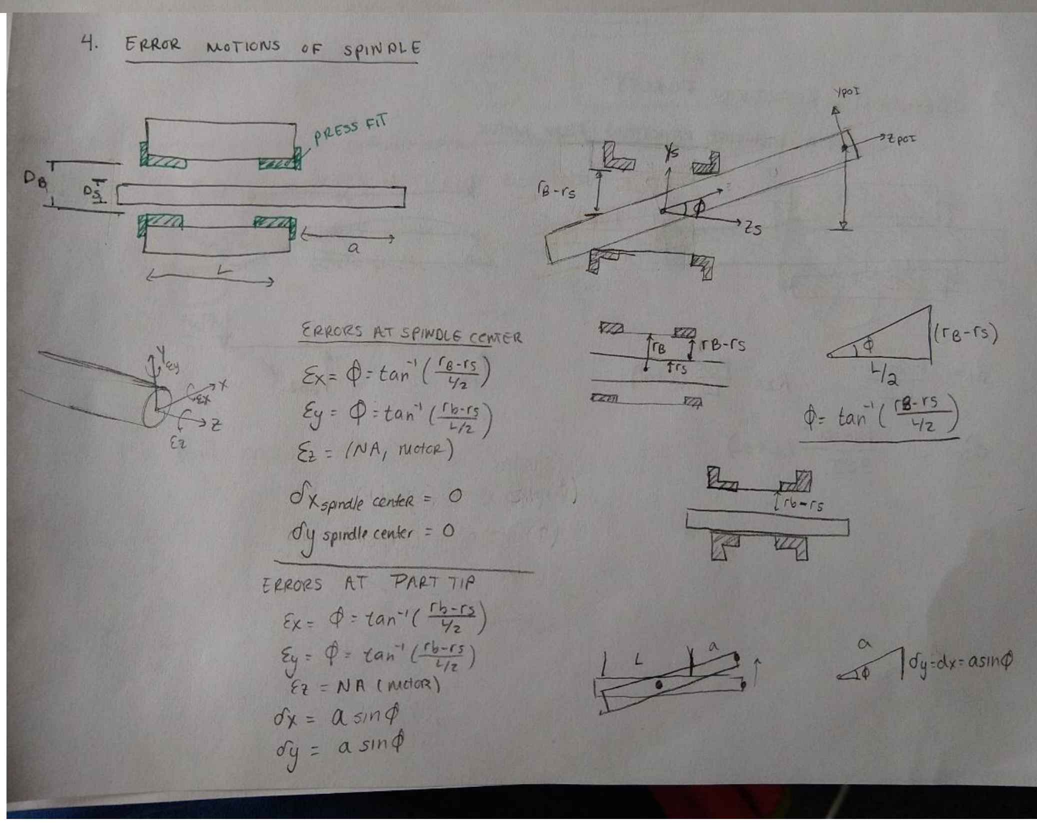


$$K_{EFF\,1} = \frac{1}{\frac{1}{K_{thrust}} + \frac{1}{K_{plain}}}$$

$$K_{EFF\,2} = \frac{1}{\frac{1}{K_{thrust}} + \frac{1}{K_{plain}} + \frac{1}{K_{spring}}}$$

$$K_{shaft} = \frac{1}{\frac{1}{K_{bolt1}} + \frac{1}{K_{flange}} + \frac{1}{K_{shaftd}} + \frac{1}{K_{bolt2}}}$$



ERRORS AT SPINDLE CENTER

$\varepsilon_x = \phi = \tan^{-1}\left(\frac{r_B - r_S}{L/2}\right)$

$\varepsilon_y = \phi = \tan^{-1}\left(\frac{r_b - r_s}{L/2}\right)$

$\varepsilon_z =$ (NA, motor)

$\delta x_{spindle\ center} = 0$

$\delta y_{spindle\ center} = 0$

ERRORS AT PART TIP

$\varepsilon_x = \phi = \tan^{-1}\left(\frac{r_b - r_s}{L/2}\right)$

$\varepsilon_y = \phi = \tan^{-1}\left(\frac{r_b - r_s}{L/2}\right)$

$\varepsilon_z =$ NA (motor)

$\delta x = a \sin\phi$

$\delta y = a \sin\phi$

$\phi = \tan^{-1}\left(\frac{r_B - r_S}{L/2}\right)$

$\delta y = dx = a \sin\phi$

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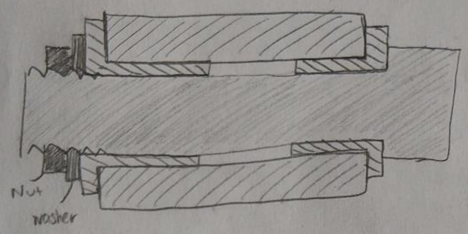
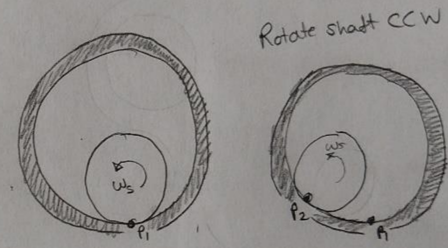
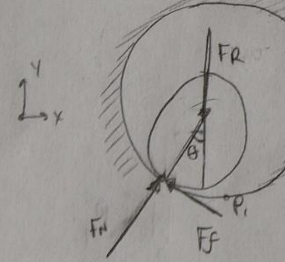
JOURNAL BEARING FRICTION



1. JOURNAL BEARING FRICTION

Rotate shaft CCW





CLIMB ANGLE θ

$$F_R = \sqrt{F_{cut}^2 + F_{thrust}^2}$$

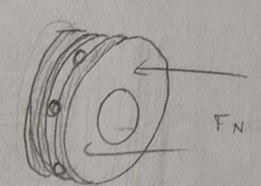
→ IF θ IS small

$$\Sigma F_y = 0 \Rightarrow F_{CUT} = F_N$$

$$F_f = F_N \mu = F_R \mu$$

$$M_f = r_{shaft} \cdot F_R \cdot \mu$$ ← FRICTIONAL TORQUE OPPOSING MOTOR TORQUE

THRUST BEARING FRICTION

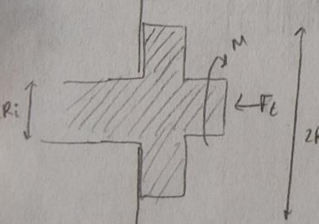


FRICTIONAL TORQUE (FRICTION DISK)

$$M = \frac{\mu F_t}{\pi (R_o^2 - R_i^2)} \int_0^{2\pi} \int_{R_i}^{R_o} r^2 \, dr \, d\theta \Rightarrow \frac{2}{3} \mu F_N \frac{R_o^3 - R_i^3}{R_o^2 - R_i^2}$$

(For a ball thrust bearing, $\mu \approx .0015$

$\mu = .175$ for plain bearing



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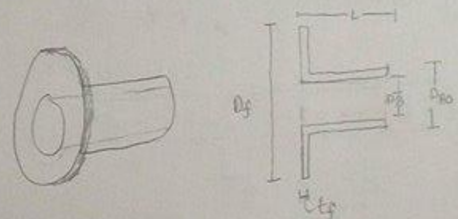
STIFFNESS MODELS

FLANGED
① Plain BEARING (IGUS J3 iglide (from sample kit))

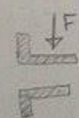


$D_f = 30mm$ $E = 2.7\times10^9$
$D_B = 20mm$
$D_{BO} = 23mm$
$t_f = 1.5mm$
$L = 21.5mm$

$r_B = D_B/2$
$r_{BO} = D_{BO}/2$
$r_f = D_f/2$

contact area

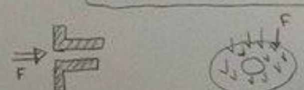
K RADIAL



Surface area of SHAFT
$A = 2\pi rh$
approximately 1/3 area will be in contact
→ can vary based on measurements

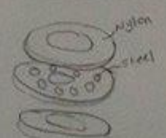
$$\boxed{K_{RADIAL} = \frac{EA}{t} = \frac{E\left(\frac{1}{3}(2\pi r_B L)\right)}{(r_{BO}-r_B)}}$$

KAXIAL

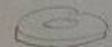


$$\boxed{K_{Axial} = \frac{EA}{t} = \frac{E(\pi)(r_f^2 - r_B^2)}{t_F}}$$

2 STIFFNESS MODEL: THRUST BEARING



Nylon

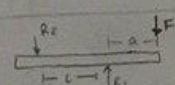


$\frac{EA}{t}$ VS

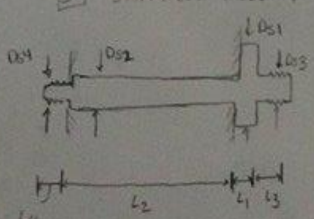
$$K_{washer} = \frac{E\pi(r_o^2 - r_i^2)}{t}$$

$$K_{ball} = \frac{F_{dynamic}}{(.01 \cdot D_{ball})}$$

3 STIFFNESS MODEL: Spindle SHAFT



$$K_{RAD} = \frac{3EI(L+a)}{a^2}$$



$$K_{axial\,1} = \frac{E_{steel}\,A_{flange}}{L_1}$$

$$K_{axial\,2} \Rightarrow \frac{E_{steel}(\pi r_2^2)}{L_2}$$

$$K_{axial\,3} = \frac{E_{steel}(\pi r_3^2)}{L_3}$$

$$K_{axial\,4} = \frac{E_{steel}(\pi r_4^2)}{L_4}$$

$$K_{shaft}^{axial} = \frac{1}{\frac{1}{K_{a1}} + \frac{1}{K_{a2}} + \frac{1}{K_{a3}} + \frac{1}{K_{a4}}}$$

PROPRIETARY INFORMATION
Witnessed And Understood By